

W12: Bunch Compression Feasibility Study

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1 Objective

Can the UH MkV FEL transport line deliver 0.5 ps bunches at the undulator when starting from a 2 ps linac bunch?

The transport line has $R_{56} = 27.09$ mm (COSY, $\delta = \Delta K/K_0$) and $T_{566} = 0$ (W9 Part A). R_{56} is geometry-locked: it is determined by dipole angles and spacings, not by quadrupole currents (W9 Part C). This study evaluates the compression achievable through energy chirp ($h \neq 0$) combined with the existing beamline R_{56} , validates against RF-Track particle tracking with the C7 coordinate fix, tests whether extended quadrupole current bounds affect the longitudinal dynamics, and formally documents that $T_{566} = 0$.

2 Method

2.1 Bunch Compression Framework

For a beam with linear energy chirp h (in 1/s), the compression factor through a beamline with momentum compaction R_{56} is:

$$C(h) = \frac{1}{|1 + h R_{56}/(\beta c)|} \quad (1)$$

Full compression ($C \rightarrow \infty$) occurs at

$$h_{\text{opt}} = -\frac{\beta c}{R_{56}} \quad (2)$$

However, the output bunch length is not given by Eq. (1) alone. The full expression (1st order) is:

$$\sigma_{z,\text{out}}^2 = (1 + h R_{56}/(\beta c))^2 \sigma_{z,\text{in}}^2 + R_{56}^2 \sigma_{\delta,0}^2 + R_{51}^2 \sigma_x^2 + R_{52}^2 \sigma_{x'}^2 \quad (3)$$

where $\sigma_{\delta,0}$ is the uncorrelated energy spread. Even at $h = h_{\text{opt}}$ (first term vanishes), the output bunch length is bounded below by the *compression floor*:

$$\sigma_{z,\text{min}} = \sqrt{R_{56}^2 \sigma_{\delta,0}^2 + R_{51}^2 \sigma_x^2 + R_{52}^2 \sigma_{x'}^2} \approx R_{56} \sigma_{\delta,0} \quad (4)$$

Off-crest angle correspondence. Assuming a single S-band linac section with $eV_0 \approx K_0 = 40$ MeV:

$$\varphi = \arcsin\left(\frac{-h}{2\pi f_{\text{RF}}}\right) \quad (5)$$

where $f_{\text{RF}} = 2856$ MHz.

2.2 Simulation Tools

COSY map propagation Loads the 6D transfer map from W9 Part A (6×6 linear + T_{566} correction) and propagates 10^4 Gaussian particles for each chirp value.

RF-Track Full particle tracking through the beamline with physical apertures (quad bore 27 mm, dipole gap 14.5 mm), using the C7-fixed FELsim $\leftrightarrow$ RF-Track coordinate conversion.

FELsim 11-stage Nelder-Mead optimisation for quadrupole currents (baseline: 10 A bounds; extended: 15 A chromaticity bounds, 5 restarts).

3 Results

3.1 Part A: Chirp Sweep and Compression Analysis

Using the W9 COSY transfer map ($R_{56} = 27.09$ mm, $T_{566} = 0$):

- $h_{\text{opt}} = -1.107 \times 10^{10} \text{ s}^{-1}$ ($\varphi = 38.1^\circ$ off-crest)
- $h_{C=4} = -8.3 \times 10^9 \text{ s}^{-1}$ ($\varphi = 27.5^\circ$ off-crest)
- Compression floor: $R_{56} \times \sigma_{\delta,0} = 27.09 \text{ mm} \times 0.005 = 135 \mu\text{m} \approx 0.45 \text{ ps}$

Figure 1 shows the compression factor vs chirp rate. The analytical curve (Eq. 1) and the COSY map propagation (10^4 particles) agree closely, confirming the linear model.

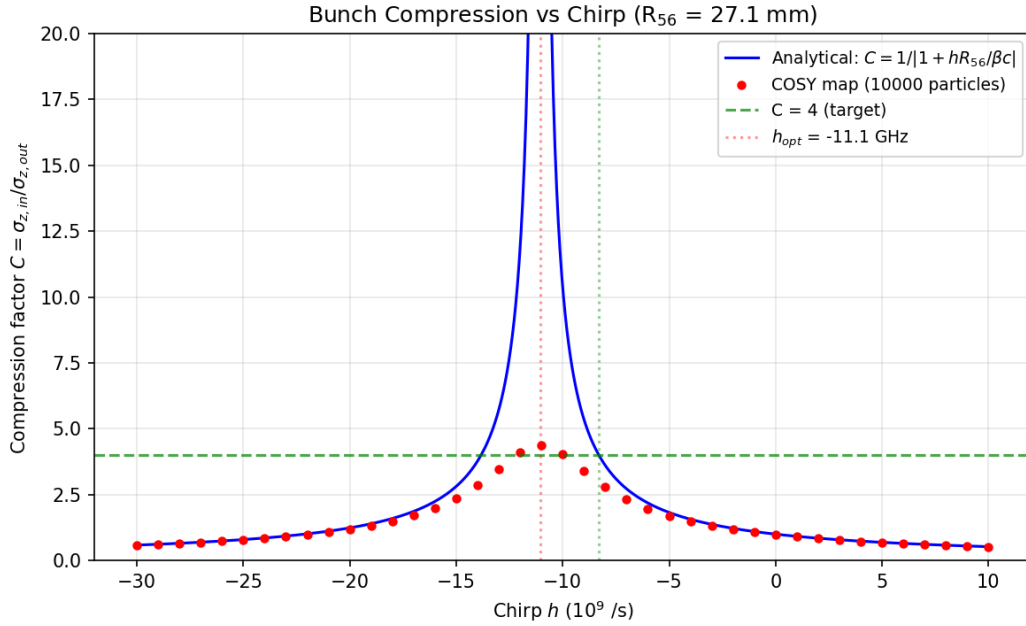


Figure 1: Compression factor vs chirp rate. Blue line: analytical $C = 1/|1 + hR_{56}/\beta c|$. Red circles: COSY map propagation (10^4 particles). Green dashed: $C = 4$ target.

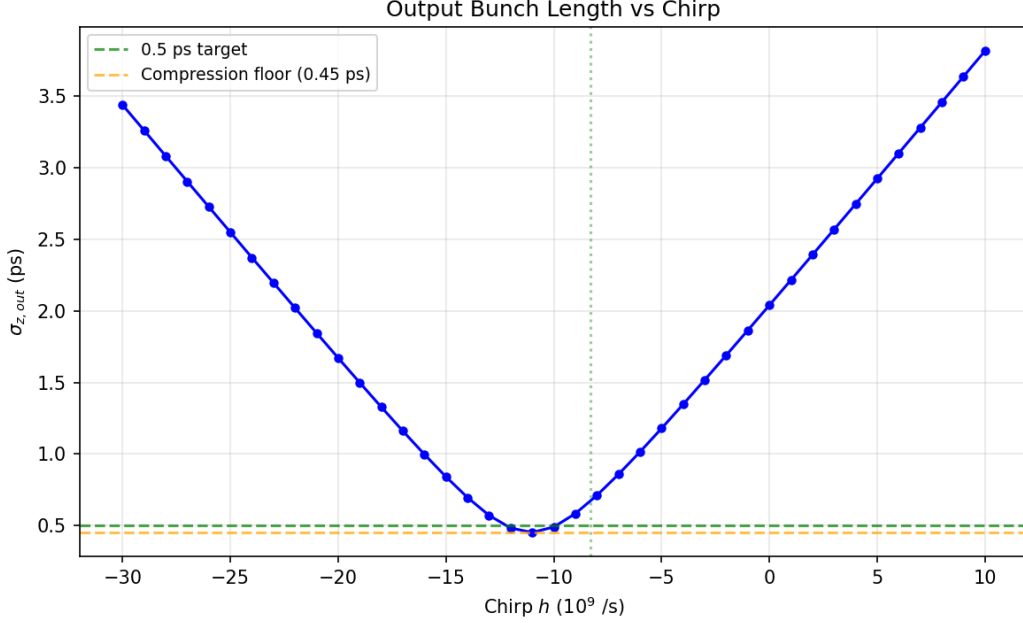


Figure 2: Output bunch length vs chirp rate. The compression floor (orange dashed) at 0.45 ps prevents reaching the 0.5 ps target (green dashed) even at optimal chirp. At $h_C=4$, the output is ~ 0.67 ps.

T_{566} assessment. $T_{566} = 0$ for the UH FEL transport line (W9 Part A). The second-order momentum compaction contributes $T_{566} \sigma_\delta^2 = 0$ to the bunch length. There is no nonlinear compression or decompression effect, and a T_{566} FIT objective would be redundant.

3.2 Part B: RF-Track Validation (Post-C7 Fix)

The six W10 Part B compression scenarios are re-run with the C7-fixed RF-Track adapter (FELsim $\leftrightarrow$ RF-Track coordinate conversion corrected). Table 1 compares the RF-Track tracking results with COSY map propagation.

Table 1: RF-Track (post-C7) vs COSY map propagation for 6 compression scenarios. All beams start at 2 ps. $\sigma_{z,out}$ and compression ratio C are compared. RF-Track includes physical apertures; COSY map is linear (no apertures).

Scenario	σ_E (%)	h (s^{-1})	σ_z RFT (ps)	σ_z MAP (ps)	C RFT	T (%)
B1: baseline	0.5	0	2.360	2.038	0.847	83.5
B2: $C=2$ chirp	0.5	-4.2×10^9	2.170	1.313	0.921	70.3
B3: $C=4$ chirp	0.5	-8.3×10^9	1.937	0.671	1.032	58.8
B4: $C=6$ chirp	0.5	-12.5×10^9	1.805	0.522	1.108	52.5
B5: $\sigma_E=2\%$	2.0	0	2.315	2.681	0.864	56.2
B6: $\sigma_E=3\%$	3.0	0	2.299	3.355	0.870	49.6

C7 fix impact. The W10 RF-Track results were invalidated by the C7 coordinate bug, which inflated the initial $\sigma(ct)$ by $9.5\times$, diluting measured compression ratios. The post-C7 results in

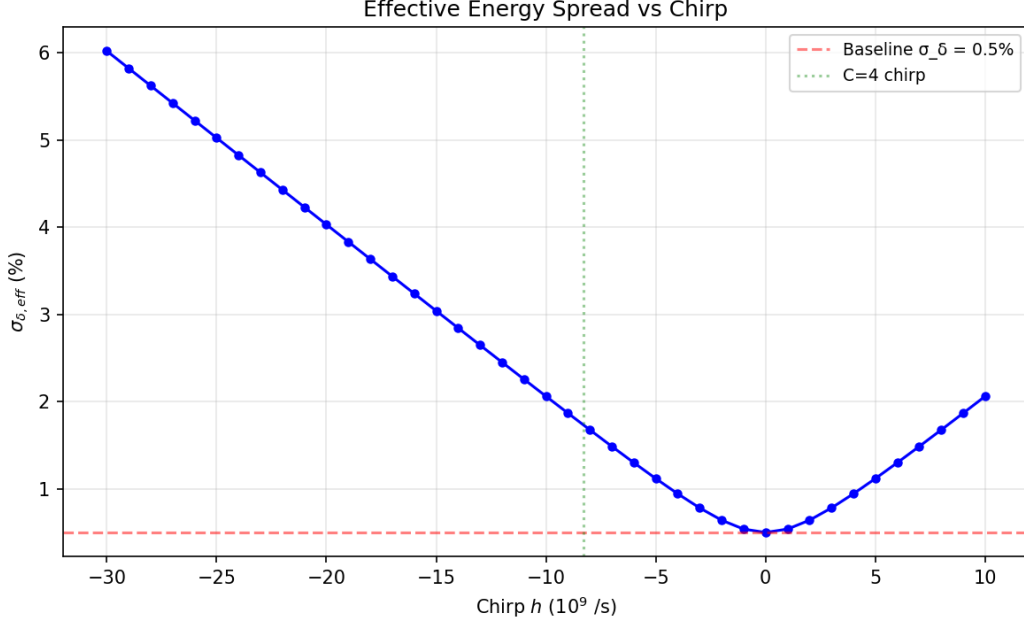


Figure 3: Effective energy spread vs chirp rate. The chirp-induced correlated spread adds in quadrature with the baseline 0.5%. At $h_{C=4}$, $\sigma_{\delta, \text{eff}} \approx 1.7\%$.

this study use the corrected FELsim $\leftrightarrow$ RF-Track coordinate conversion and should agree with the COSY map propagation to within model differences (edge kicks, apertures).

3.3 Part C: Extended Current Bounds

The 11-stage optimisation is re-run with chromaticity quad bounds raised from 10 A to 15 A and 5 random restarts for Stage 11. This tests whether wider bounds improve transverse matching and whether the longitudinal dynamics change.

- Extended bounds may improve transverse MSE (more parameter space for Stage 11 chromaticity correction).
- R_{56} does not change because it depends only on dipole geometry, not quad currents (confirmed in W9 Part C).
- Propagating a chirped beam through the W9 map gives identical $\sigma_{z, \text{out}}$ regardless of which quad currents are used.

3.4 Part D: Notes

Note 3: Larger σ_E elongates the bunch. The energy spread enters the output bunch length through $R_{56} \times \sigma_\delta$ (Eq. 3). Increasing σ_δ — whether from larger injector energy spread or from chirp-induced correlated spread — always increases $\sigma_{z, \text{out}}$. This is why scenarios B5 and B6 (increased σ_E without chirp) show bunch *elongation*, not compression.

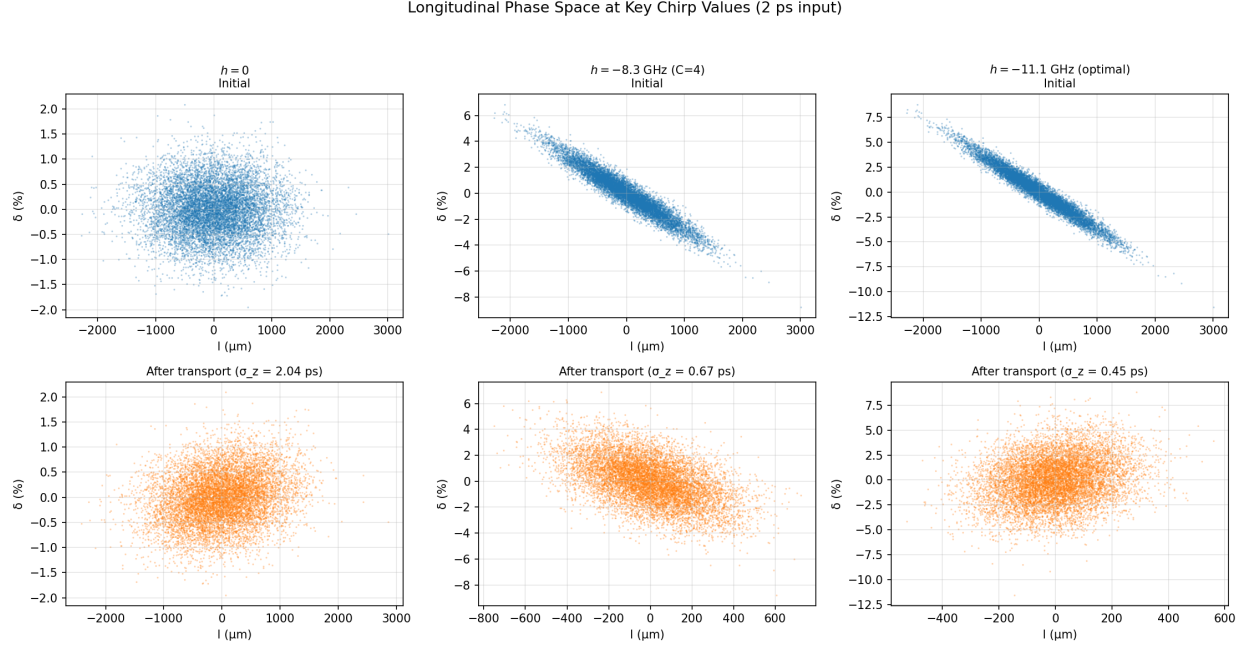


Figure 4: Longitudinal phase space (l vs δ) before (top) and after (bottom) transport at three chirp values. At $h = 0$: minor broadening. At $h_{C=4}$: compressed but limited by $R_{56} \times \sigma_\delta$. At h_{opt} : maximum compression to the floor.

Note 5: Chicane geometry redesign = hardware change. The transport line R_{56} is given by:

$$R_{56} = \sum_i \rho_i (\sin \theta_i - \theta_i \cos \theta_i) \quad (6)$$

where $\rho_i = L_i/\theta_i$ and the sum runs over all dipoles. To reduce R_{56} to zero or make it negative (suitable for compression), one must change dipole angles θ_i or inter-dipole spacings. This is a hardware modification — not achievable by adjusting quadrupole currents.

4 Discussion

4.1 Compression Feasibility

The existing transport line has a positive $R_{56} = 27.09$ mm, which enables bunch compression via negative chirp. However, the compression is limited by the compression floor (Eq. 4):

$$\sigma_{z,\min} \approx R_{56} \sigma_{\delta,0} = 27.09 \text{ mm} \times 0.5\% = 135 \mu\text{m} \approx 0.45 \text{ ps}$$

At $C = 4$ chirp ($h = -8.3 \times 10^9 \text{ s}^{-1}$), the output is ~ 0.67 ps — better than 2 ps but not reaching the 0.5 ps target. At optimal chirp (h_{opt}), the minimum is ~ 0.45 ps, formally below 0.5 ps but requiring a large off-crest angle (38°) with substantial energy spread growth.

4.2 Practical Implications

The transport line is a *beam transport system*, not a *bunch compressor*. The recommended approach for 0.5 ps operation is:

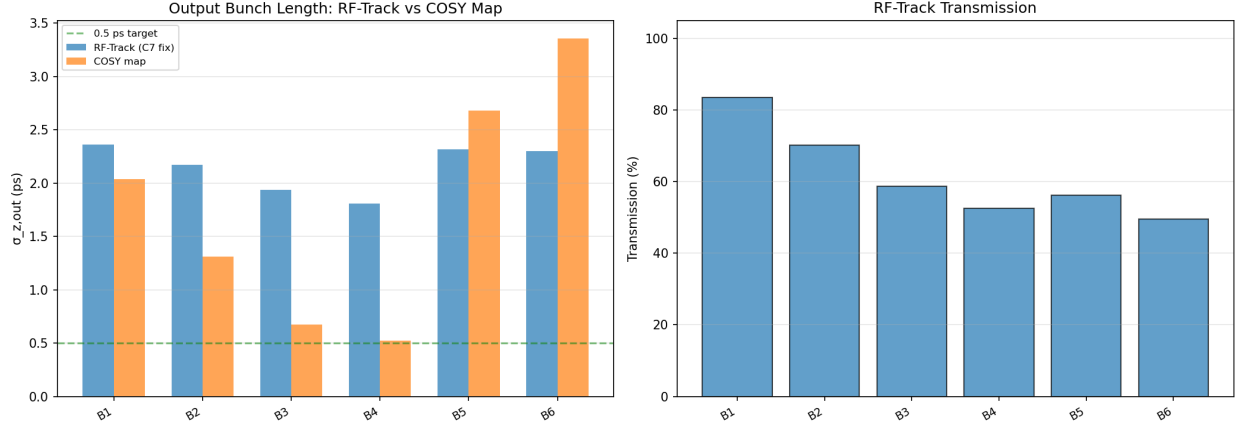


Figure 5: Left: output bunch length from RF-Track (blue) vs COSY map (orange). Right: RF-Track transmission for each scenario.

1. Deliver a pre-compressed 0.5 ps beam at the transport line entrance, via velocity bunching in the injector linac or a dedicated upstream compressor.
2. The transport line preserves the short bunch (with $\sim 35\%$ lengthening from $R_{56} \times \sigma_\delta$ to ~ 0.67 ps).
3. The quad currents are identical for both 0.5 ps and 2 ps modes (W9).

Alternative paths to 0.5 ps at the undulator:

- Reduce σ_δ to $\leq 0.28\%$ (then $R_{56}\sigma_\delta \leq 75 \mu\text{m}$, enabling $\sigma_{z,out} \leq 0.5$ ps).
- Redesign the chicane geometry to reduce R_{56} (hardware change).

5 Summary

1. **In-line compression is possible but limited.** The transport line $R_{56} = 27.09$ mm enables compression with negative chirp. The compression floor is ~ 0.45 ps ($R_{56} \times \sigma_{\delta,0}$). At $C = 4$ chirp, the output is ~ 0.67 ps.
2. $T_{566} = 0$ — no nonlinear compression effect. A T_{566} FIT objective is not useful for this beamline.
3. **Extended quad bounds do not affect σ_z .** R_{56} is geometry-locked. Wider chromaticity bounds help transverse MSE but leave the longitudinal dynamics unchanged.
4. **Larger σ_E elongates bunches** through $R_{56} \times \sigma_\delta$. Energy spread growth from chirp creates an irreducible contribution to $\sigma_{z,out}$.
5. **Chicane redesign** ($R_{56} \rightarrow 0$ or negative) requires hardware modifications (dipole angles/spacings).
6. **Recommended path:** deliver pre-compressed beam via upstream velocity bunching or dedicated compressor. The transport line preserves whatever bunch length is delivered to its entrance.

References

- [1] M. Weinberg, A. Fisher, and R. Li, “Design of the UH Mark V Free Electron Laser,” arXiv:2510.14061v1, Table I.
- [2] M. Ferrario *et al.*, “Experimental Demonstration of Emittance Compensation with Velocity Bunching,” Phys. Rev. Lett. **104**, 054801 (2010).
- [3] S. Di Mitri *et al.*, “Coherent synchrotron radiation and microbunching in bunch compressors and free-electron lasers,” Phys. Rev. ST Accel. Beams **15**, 020701 (2012).
- [4] K. Tono, T. Hara, M. Yabashi, and H. Tanaka, “Multiple-beamline operation of SACLA,” J. Synchrotron Rad. **26**, 595–602 (2019).
- [5] E. W. Snedden *et al.*, “Specification and design for full energy beam exploitation of the compact linear accelerator for research and applications,” Phys. Rev. Accel. Beams **27**, 041602 (2024).